CHAPTER 10

Trigonometry: circle, radians and angles associated

Chapter outline

Objectives. Convert degrees to radians; plot points; calculate sines and cosines of standard angles; use the symmetries of the circle.
Prerequisites. Circle, Pythagoras, cosine and sine in a right triangle, symmetries, fractions.
Annual timeline. 2·11 March: 8 h

Diagnostic check: six questions, 10 minutes

Answer without looking at your notes. Indicate the questions for which help is needed. This diagnostic test is not graded.

1. Circumference of a circle of radius 1?

2. How many degrees is a quarter turn?

3. Calculate $\sqrt{\frac{1}{2}}$.

4. Coordinates of the reflection of ($a; b$) in the x-axis?

5. Simplify $\frac{5\pi}{3} - 2\pi$.

6. In a right triangle, opposite side 3 and hypotenuse 5: sine?

After the diagnostic check. Go over the flagged concepts with the teacher; redo a similar question, then explain the rule used.

Oral work and logic. Present an equality between associated angles, naming the symmetry that justifies it; explain what a radian measures.

Investigate before formalising

Construct, without a protractor, a point at angle $\frac{\pi}{3}$ then a point at angle $\frac{\pi}{4}$. Justify the exact values of their coordinates from the figures.

Organisation. Individual work 5 min, discussion in pairs 5 min, whole-class sharing 5 min. Keep a record of an attempt, even an unfinished one. A conjecture must then be justified.

10.1 Measuring an angle in radians

Definition and results

The unit circle is the circle with centre O and radius 1, oriented anticlockwise. A radian corresponds to an arc of length 1 on this circle. One turn is 2 π radians, half a turn π. On a circle of radius r, an arc length is $r\theta$ for a positive angle θ expressed in radians.

Status and justification. Definitions and proportionality between angle at the centre and arc length. The conversion is 180 $^\circ = \pi$ radians; multiply the degrees by $\frac{\pi}{180}$.

Worked example. $60^\circ = \frac{\pi}{3}$; $\frac{3\pi}{2} = 270^\circ$. An angle of $\frac{\pi}{3}$ on a circle of radius 6 cm gives an arc of 2 π cm.

Watch out

The length of an arc is not that of the chord. In trigonometric calculations, check whether the tool is in degree or radian mode.

Immediate application. Convert 45 $^\circ$, 120° ; give the arc with radius 2 m and angle $\frac{3\pi}{4}$.

10.2 Wrapping and the image of a real number

Definition and results

To each real number x corresponds a point M on the circle obtained by a directed path of length x from I(1;0). Two real numbers have the same image if their difference is an integer multiple of 2 π. A negative number indicates travel in the clockwise direction.

Status and justification. Adding or removing a whole number of turns keeps the end point. Conversely, returning to the same point requires a whole number of turns.

Worked example. $\frac{13\pi}{3} = 4\pi + \frac{\pi}{3}$ has the same image as $\frac{\pi}{3}$. $-\frac{\pi}{2}$ has the same image as $\frac{3\pi}{2}$: point $(0; -1)$.

Watch out

Add π gives the opposite point, not the same point.

Immediate application. Plot the images of $\frac{7\pi}{2}$ & $-\frac{5\pi}{3}$ by reducing them to one turn.

10.3 Sine, cosine and standard values

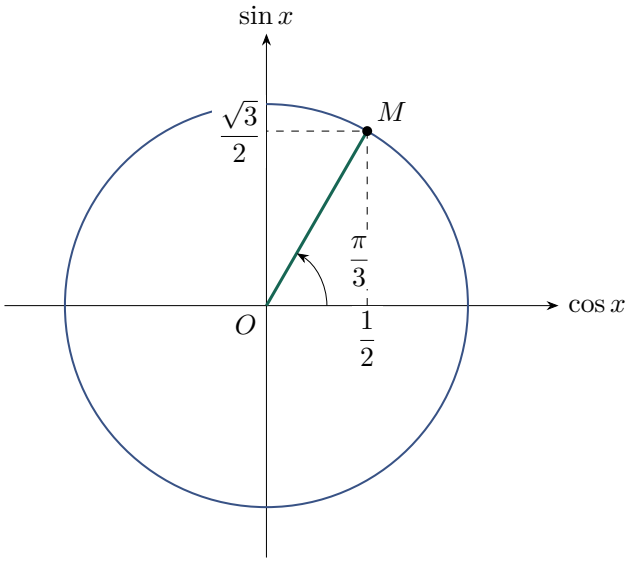
Definition and results

If M is the image of x, $M = (\cos x; \sin x)$. Hence $-1 \leq \cos x, \sin x \leq 1$ and $\cos^2 x + \sin^2 x = 1$.

For $x = 0, \frac{\pi}{6}, \frac{\pi}{4}, \frac{\pi}{3}, \frac{\pi}{2}$, the cosines are 1 $, \frac{\sqrt{3}}{2}, \frac{\sqrt{2}}{2}, \frac{1}{2}, 0$ and the sines 0 $, \frac{1}{2}, \frac{\sqrt{2}}{2}, \frac{\sqrt{3}}{2}, 1$.

Status and justification. À $\frac{\pi}{4}$, the two coordinates are equal and positive; Pythagoras gives $2a^2 = 1$, therefore $a = \frac{\sqrt{2}}{2}$. An equilateral triangle with side 1, cut by its altitude, has half-base $\frac{1}{2}$ and height $\frac{\sqrt{3}}{2}$: $\cos(\frac{\pi}{3}) = \frac{1}{2}$, $\sin(\frac{\pi}{3}) = \frac{\sqrt{3}}{2}$. The complementary angle gives $\frac{\pi}{6}$.

Worked example. The point at angle $\frac{\pi}{3}$ is $(\frac{1}{2}; \frac{\sqrt{3}}{2})$. The scalar product of two unit vectors making this angle equals its cosine, so $\frac{1}{2}$. This links the geometry chapters to the circle.



For the angle $\frac{\pi}{3}$, the projections of M give its cosine and sine directly. The fractions are placed outside the lines of the drawing.

Watch out

The cosine is the x-coordinate, the sine the y-coordinate. The identity gives an absolute value: the quadrant determines the sign.

Immediate application. If $\cos x = \frac{3}{5}$ and the image of x is in the fourth quadrant, give $\sin x$.

10.4 Related angles and symmetries

Definition and results

The symmetries of the circle give $\cos(-x) = \cos x$, $\sin(-x) = -\sin x$; $\cos(\pi - x) = -\cos x$, $\sin(\pi - x) = \sin x$; $\cos(\pi + x) = -\cos x$, $\sin(\pi + x) = -\sin x$. Adding 2 $k\pi$, k integer, does not change neither sine nor cosine.

Status and justification. Each equality can be read from the coordinates of the symmetric points: the horizontal axis for $-x$, vertical axis for $\pi - x$, centre O for $\pi + x$. This is not a list of formulas without justification.

Worked example. $\cos(\frac{5\pi}{6}) = -\frac{\sqrt{3}}{2}$, $\sin(\frac{5\pi}{6}) = \frac{1}{2}$ by symmetry of $\frac{\pi}{6}$. $\cos(-\frac{\pi}{4}) = \frac{\sqrt{2}}{2}$, $\sin(-\frac{\pi}{4}) = -\frac{\sqrt{2}}{2}$.

Watch out

Two angles with the same cosine do not necessarily have the same sine. Reduce the angle, then read the signs.

Immediate application. Calculate the two coordinates of the images of $\frac{4\pi}{3}$ and $\frac{7\pi}{4}$.

10.5 Approximating pi and reusing ideas

Definition and results

A regular n-sided polygon inscribed in the unit circle has semi-perimeter $n\sin(\frac{\pi}{n})$; a polygon has semi-perimeter $n\tan(\frac{\pi}{n})$, with $\tan\theta = \frac{\sin\theta}{\cos}\theta$ here. Archimedes' method doubles the number of sides and calculates the new lengths by Pythagoras, without knowing pi in advance.

Status and justification. The perimeter of the inscribed polygon is less than that of the circle, and that of the circumscribed one is greater: geometric bounds assumed. If c is the chord of one side, set $h = \sqrt{1 - \frac{c^2}{4}}$; the chord of the half-angle is $\frac{c}{\sqrt{2(1+h)}}$. This formula comes from Pythagoras and avoids subtracting two nearly equal numbers.

Worked example. For the inscribed hexagon, c=1 and semi-perimeter 3. The circumscribed one gives 2 $\sqrt{3} \approx$ 3,4641. Doubling the number of sides tightens the bounds. The given program starts from n=6 and c=1.

Watch out

Using a sine function directly with pi already known would check a calculation, but would not construct an independent approximation of pi.

Immediate application. For n=6 and c=1, calculate the new chord after doubling.

Exercise path

Start with exercises 1 and 2. Then choose exercise 3 or 4 without help. Exercises 5 to 7 are a bank of transfer, investigation or programming tasks: the teacher assigns one. Exercise 8 is revisited later.

Exercise 10.1 · Convert and Measure

The unit circle is used to measure angles in radians.

1. Convert 150 ° & $-90°$ into radians, showing the conversion factor.
2. A circular sector has a radius of 4 cm and an angle of $\frac{\pi}{3}$ radians. Calculate the exact length of its arc.
3. Explain why an angle measure can be negative while a geometric arc length is positive.

Exercise 10.2 · Reduce angles

Consider the real numbers $\frac{17\pi}{6}$, $-\frac{7\pi}{4}$ and 5π.

1. For each, determine a representative in [0; 2 π[having the same image on the unit circle. Show the number of turns added or removed.
2. Plot the three images on a circle and label them with the original real numbers.

Exercise 10.3 · Exact values

We look for the exact values of the cosine and sine of $\frac{2\pi}{3}$, $-\frac{\pi}{6}$ and $\frac{5\pi}{4}$.

1. Place the three images on the unit circle. For each, state the expected sign of its two coordinates.
2. Calculate the exact values using a standard angle and a symmetry. Present the answer in a table: angle, cosine, sine.

Exercise 10.4 · Find the sign

A real x satisfies $\sin x = \frac{5}{13}$. Its image on the unit circle lies in the second quadrant, hence to the left of the y-axis and above the x-axis.

1. Determine the exact value of cos x. Justify both its absolute value and its sign.

2. Explain why knowing only sin x would not have been enough to choose the sign of cos x.

Exercise 10.5 · A wheel

A wheel of radius 0 ,30 m rolls on horizontal ground without slipping. It turns through an angle $\frac{5\pi}{3}$ radians. We want the distance travelled by its axle in the direction of motion.

1. Calculate this distance exactly, explaining the link with the unrolled arc length.

2. Explain why the length of a chord of the circle must not be used. Is the path of a fixed point on the rim of the wheel, seen from the ground, the same as that of the axle?

Exercise 10.6 · Chord and scalar product

On the unit circle with centre O, the point A is the image of 0 and the point B that of $\frac{2\pi}{3}$.

1. Determine the exact coordinates of A and B, then calculate the length AB.

2. Find this length again by applying Al-Kashi's formula to the triangle OAB. State the lengths and the angle used.

Exercise 10.7 · Read an equation on the circle

We look for the real numbers x in the interval [0; 2 π[whose image on the unit circle has x-coordinate $\frac{1}{2}$.

1. Express the condition on the x-coordinate as an equation involving cos x.

2. Draw the corresponding vertical line on a circle, locate its points of intersection and give the two solutions. Check that they belong to the required interval.

Exercise 10.8 · Delayed review

Consider the angle with measure $\frac{25\pi}{4}$ radians. Answers must be exact, without decimal approx.

1. Reduce this angle by removing a whole number of complete turns.

2. Deduce its cosine and sine. Justify the values using the unit circle.

Independent work. Do only the task assigned at the end of the session, as set in the dated scheme of work: 10-20 min maximum. The exercises depend on the concepts already studied; some questions are therefore postponed. Revisit exercise 8 one week after the chapter (10 min). If the planned time is exceeded, note the difficulty and bring your attempts; do not try to do the whole reserve.

To be done in the CQFD textbook

These exercises are done in your textbook. The scheme of work gives the session; they replace the homework planned for that day (10 to 20 minutes in total).

CQFD, exercise 32 p. 220, questions a and b. Use the exact value given and the associated angles, without a calculator.

CQFD, Exercise 39 p. 220, Questions 1, 2, 3 and 5. Do the four questions on calculation and reading the circle; justify the answers on your script.

MCQ 1: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. $180°$ is
   - A. 2π
   - B. π
   - C. $\frac{\pi}{2}$
   - D. 1
2. $\cos(\frac{\pi}{3}) =$
   - A. $\frac{1}{2}$
   - B. $\frac{\sqrt{3}}{2}$
   - C. 0
   - D. 1
3. $\sin(-\frac{\pi}{2}) =$
   - A. 1
   - B. 0
   - C. $\frac{1}{2}$
   - D. -1
4. $\cos^2 x + \sin^2 x =$
   - A. 0
   - B. 2
   - C. 1
   - D. x
5. The same point corresponds to angles differing by
   - A. π
   - B. $2k\pi$, k an integer
   - C. $\frac{\pi}{2}$
   - D. 1

MCQ 2: understand and choose

Five teaching questions with a single answer. Choose one answer per question; then explain one distractor. This chapter format does not reproduce the official format of the EAM.

1. $\cos(\pi - x) =$

   A. $\cos x$

   B. $\sin x$

   C. $-\cos x$

   D. $-\sin x$

2. $\sin(\frac{\pi}{4}) =$

   A. $\frac{\sqrt{2}}{2}$

   B. $\frac{1}{2}$

   C. $\frac{\sqrt{3}}{2}$

   D. $\sqrt{2}$

3. Image of $\frac{3\pi}{2}$:

   A. $(1;0)$

   B. $(-1;0)$

   C. $(0;1)$

   D. $(0;-1)$

4. Arc with radius 3 and angle 2 radians: length

   A. $\frac{2}{3}$

   B. 6

   C. $\frac{3}{2}$

   D. 6π

5. Inside the second quadrant: sign of the cosine

   A. positive

   B. zero everywhere

   C. negative

   D. cannot be determined

True or false? Justify.

1. $\cos(-x) = -\cos x$ for all x.

2. $\sin(x + 2\pi) = \sin x$.

3. Knowing sin x always determines cos x.

Personal review

I reduce the angles by complete turns; I read cosine and sine in the right order; I justify the signs; I keep the exact values.
For each objective, indicate: mastered without help; achieved with help; needs more work. Write an example and a mistake to avoid.

Solutions and teaching notes, chapter 10

Teacher use. The following answers concern the tasks not solved in the lesson. First have pupils make their approach explicit; provide graduated help before the solution. The durations are estimates to be calibrated in class.

Diagnostic and remediation

1. 2π. *Action:* Link radius, diameter and perimeter.
2. $90°$. *Action:* Divide a circle into four sectors.
3. $\dfrac{\sqrt{2}}{2}$. *Action:* Rationalise and check the square.
4. $(a;-b)$. *Action:* Plot both points.
5. $-\dfrac{\pi}{3}$. *Action:* Put over a common denominator.
6. $\dfrac{3}{5}$. *Action:* Identify the reference angle.

Investigation

Equilateral triangle then right-angled isosceles triangle. Pythagoras gives the height $\dfrac{\sqrt{3}}{2}$ and equal coordinates $\dfrac{\sqrt{2}}{2}$. The signs are positive in the first quadrant.

Direct applications

1. $\dfrac{\pi}{4}$, $\dfrac{2\pi}{3}$; length 3 $\dfrac{\pi}{2}$ m.
2. $\dfrac{7\pi}{2}$ has the image of $3\dfrac{\pi}{2}$; $-\dfrac{5\pi}{3}$ that of $\dfrac{\pi}{3}$.
3. $\sin^2 x = \dfrac{16}{25}$; the sine is negative, so $-\dfrac{4}{5}$.
4. $(-\dfrac{1}{2};-\dfrac{\sqrt{3}}{2})$ and $(\dfrac{\sqrt{2}}{2};-\dfrac{\sqrt{2}}{2})$.
5. $h = \dfrac{\sqrt{3}}{2}$; new chord $\dfrac{1}{\sqrt{2+\sqrt{3}}} = \sqrt{2-\sqrt{3}}$.

Exercises: methods and answers

Exercise 10.1 · Convert and measure. $\dfrac{5\pi}{6}$ & $-\dfrac{\pi}{2}$; arc $\dfrac{4\pi}{3}$ cm. The negative sign indicates a orientation; a geometric length is positive.

Exercise 10.2 · Reduce angles. $\dfrac{5\pi}{6}$, $\dfrac{\pi}{4}$, π. Remove or add multiples of 2 π.

Exercise 10.3 · Exact values. $(-\frac{1}{2};\frac{\sqrt{3}}{2})$; $(\frac{\sqrt{3}}{2};-\frac{1}{2})$; $(-\frac{\sqrt{2}}{2};-\frac{\sqrt{2}}{2})$, in the form (cosine ; sine).

Exercise 10.4 · Find the sign. $\cos^2 x = 1-\frac{25}{169}=\frac{144}{169}$; negative cosine, so $-\frac{12}{13}$.

Exercise 10.5 · A wheel. Distance $r\theta = 0{,}3(\frac{5\pi}{3})=\frac{\pi}{2}$ m. This corresponds to unrolling the arc; the chord joins two positions on the circle and does not have this length. A point fixed on the wheel traces a different path relative to the ground.

Exercise 10.6 · Chord and scalar product. $A(1;0)$, $B(-\frac{1}{2};\frac{\sqrt{3}}{2})$; $AB^2=\frac{9}{4}+\frac{3}{4}$ = 3. Al-Kashi : $1+1-2\cos(\frac{2\pi}{3})$ = 3. Hence $AB=\sqrt{3}$.

Exercise 10.7 · Read an equation on the circle. The vertical line with x-coordinate $\frac{1}{2}$ intersects the circle at the angles $\frac{\pi}{3}$ and $\frac{5\pi}{3}$. The two points have opposite sines.

Exercise 10.8 · Delayed review. $\frac{25\pi}{4}=6\pi+\frac{\pi}{4}$. Three full turns do not change the image, so both values are equal to $\frac{\sqrt{2}}{2}$.

CQFD, exercise 32 p. 220, questions a and b. a : $\cos(\frac{6\pi}{5})=\cos(\pi+\frac{\pi}{5})=-\frac{1+\sqrt{5}}{4}$. b : $\cos(-\frac{\pi}{5})=\cos(\frac{\pi}{5})=\frac{1+\sqrt{5}}{4}$, since cosine is even.

CQFD, Exercise 39 p. 220, Questions 1, 2, 3 and 5. 1 : $\frac{\pi}{6}+\frac{\pi}{3}=\frac{\pi}{2}$. 2 : yes, $\cos(\frac{\pi}{3})=\frac{1}{2}=0{,}5$ is a decimal. 3: the principal measure in] $-\pi;\pi$] is $-\frac{5\pi}{6}$. 5 : $\sin(\frac{4\pi}{3})=-\frac{\sqrt{3}}{2}$.

MCQ: answers and reasons

MCQ 1.

1. B. Half-turn.

2. A. Equilateral triangle.

3. D. Lowest point of the circle.

4. C. Equation of the unit circle.

5. B. Complete turns.

MCQ 2.

1. C. Vertical symmetry.

2. A. Isosceles right triangle.

3. D. Three-quarters of a turn.

4. B. $r\theta = 6$.

5. C. Negative x-coordinate inside the quadrant.

True or false

1. False: at 0, the two sides would be 1 and -1. Cosine is even.

2. True: a complete turn keeps the image.

3. False: $\frac{\pi}{6}$ and $\frac{5\pi}{6}$ have the same sine but opposite cosines.

Differentiated teaching. If a pupil is stuck on a technical point, suggest only the first step (domain, factor or reference sample space), then remove the help. For an independent student, ask for a second method and the handling of a boundary case. Orally, allow an initial rephrasing in the language the pupil is fluent in, then require a precise mathematical sentence in French.